

PARADI JAGADEESWARARAO
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SUMMARY:

Results-driven RF Engineer with a strong background in testing, design, and development of electronic and telecommunication systems. Detail-oriented and highly organized professional with proven ability to manage multiple projects efficiently and meet tight deadlines. Adept at troubleshooting complex technical issues, optimizing system performance, and supporting cross-functional engineering teams. Recognized for innovation, problem-solving, and effective collaboration in achieving organizational goals. Skilled in systems analysis, project management, and technical testing, with a commitment to delivering high-quality engineering solutions that drive operational success.

Professional Experience :

FOXCONN HON HAI TECHNOLOGY INDIA MEGA DEV.PRIVATE LIMITED, Bengaluru.

RF Engineer

Nov 2023 – Present

- Conduct detailed Failure Analysis (FA) and prepare yield reports; review findings with customers and present results in technical meetings.
 - Possess a strong understanding of antenna design, performance optimization, and RF band test failure analysis.
 - Perform process capability (Cp, Cpk) studies to support data-driven decision-making and improve production consistency.
 - Utilize advanced communication testers for 5G, Wi-Fi 6E, and Bluetooth performance testing and validation.
 - Execute Design of Experiments (DOE) based on customer priorities to optimize testing and process parameters.
 - Analyze top issues and drive initiatives for rejection and scrap reduction through collaboration with maintenance, manufacturing, and quality teams.
 - Conduct tester correlation studies and impedance checks to ensure signal integrity and testing accuracy.
 - Lead failure analysis and process improvement initiatives using 4M and 5W1H methodologies; perform detailed tear-down studies to identify root causes.
 - Proficient in data analytics, report preparation, and the use of electronic measurement tools (CT, VHX, Microscope, Multimeter).
 - Apply in-depth knowledge of RF and Access Point (AP) technologies to identify and resolve complex hardware-related issues.
 - Conduct training sessions for junior engineers on equipment operation, testing procedures, and safety protocols to enhance team capability.
 - Manage project timelines and resources, ensuring on-time and on-budget completion of engineering deliverables.
 - Collaborate with cross-functional teams to analyze system failures and develop preventive maintenance strategies, improving reliability and uptime.
 - Utilize CAD software to design and document technical drawings and specifications compliant with industry standards.
 - Lead troubleshooting and root cause analysis on complex systems, implementing corrective actions to prevent recurrence.
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Debug Engineer

Nov 2022 - Sep 2023

- Performed testing and debugging of 5G radio units (Air 3219 and Air 3268) to ensure compliance with performance and quality standards.
 - Conducted PCB-level and unit-level testing, identifying faults and implementing corrective actions to enhance reliability.
 - Possess working knowledge of OTA (Over-The-Air) and MOB (Mobile Performance Testing) methodologies to validate RF performance and communication stability.
 - Utilized RF test instruments such as spectrum analyzers, network analyzers, and signal generators for diagnosing complex circuit issues.
 - Collaborated with R&D and design teams to analyze test data and support hardware validation activities.
 - Assisted in root cause analysis of test failures and contributed to continuous improvement of test procedures.
 - Prepared detailed test reports and debug logs, ensuring traceability and alignment with customer and internal quality requirements.
 - Supported prototype evaluations and new product introductions (NPI) by verifying design functionality and ensuring production readiness.
 - Ensured adherence to ESD, safety, and quality standards during testing and debug operations.
 - Demonstrated strong problem-solving and analytical skills in resolving RF signal and hardware-related issues efficiently.
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IMEDS GLOBAL PRIVATE LIMITED, Visakhapatnam

Hardware Test Engineer

Feb 2022 - Oct 2022

- Contributed to the research and development (R&D) of a 10 LPM Oxygen Concentrator, focusing on hardware testing and validation of control systems.
 - Involved in the hardware design and testing of control cards, ensuring optimal functionality and reliability of electronic circuits.
 - Supported prototype development and functional verification, collaborating closely with design and production teams to refine system performance.
 - Conducted performance evaluation and failure analysis of electronic components under various environmental and load conditions.
 - Prepared test reports, design documentation, and technical presentations to support design reviews and product certifications.
 - Utilized electronic measurement tools (multimeter, oscilloscope, power analyzer) for circuit validation and troubleshooting.
 - Participated in design improvements and circuit optimization, helping to enhance device efficiency and reliability.
 - Ensured compliance with medical device safety and regulatory standards, maintaining strict quality and documentation procedures.
 - Assisted in the integration and calibration of sensors, valves, and flow systems used in oxygen concentrators.
 - Collaborated with cross-functional teams to achieve project milestones within deadlines, demonstrating strong technical and communication skills.
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- Led RF/microwave design and development at both module and system levels, delivering robust transmit/receive solutions across product life cycles.
- Performed PCB layout review with EMI/EMC mitigation strategies to reduce emissions and improve immunity during the PCB design phase.
- Designed and validated product-level RF subsystems including LNAs, HPAs, frequency synthesizers, mixers, power splitters, filters, up/down-converters, and PLL/DDS architectures.
- Executed module-level development and testing: cascade analysis, EVM measurement for modulators, receiver design verification, noise-figure calculations, and link power-budget analysis.
- Designed transceivers for digital modulation schemes (BPSK, QPSK) and performed reliability and performance analyses under varied signal conditions.
- Developed wideband RF reception and transmission elements (operating ~100 MHz – 3 GHz) — including high dynamic range low-noise acquisition, filtering, and frequency translation circuits.
- Designed and tuned power amplifiers with careful impedance matching, and validated performance for gain, spurious, harmonics, and phase noise.
- Investigated and resolved EMI/EMC issues discovered during test; implemented corrective design changes and test process improvements.
- Contributed to development of ruggedized systems meeting airworthiness requirements for Indian Navy applications (aircraft, ship and ground installations).
- Prepared and maintained engineering and test documentation: ATP, QTP, QAP, ETP, TM, UM, and product PDFs to support manufacturing, qualification, and handover.
- Coordinated with cross-functional teams (mechanical, firmware, test, quality) and external vendors to ensure design for manufacturability and supply readiness.

Selected projects & achievements

- MSS TRE M-II (S-Band) — Designed RF front-end and a multi-stage PA chain (driver → final) delivering 40 W output for voice/data communications at multiple data rates; supported defense certification activities.
- PA Module Band-III (Repeater Tx section) — Engineered pre-amp, driver and final stages to achieve 20 W transmit output for voice/data repeater applications; optimized linearity and thermal performance.
- MSS Aircraft Tracking System — Implemented RF transmitter chain for GPS telemetry uplink to GSAT-6 HUB; ensured continuous 2-second coordinate updates and produced a 20 W PA for airborne terminal.
- LPJ Power Amplifiers (Low-Power Jammers) — Tuned and validated pre-amp and driver stages for jamming applications; focused on stability, spectral containment and thermal robustness.

Additional contributions

- Performed root-cause analysis during field and lab failures and proposed corrective design/process actions.

- Supported qualification testing (environmental, vibration, thermal) and implemented design changes to improve MTBF and operational reliability.
- Mentored junior engineers through design reviews and test plan execution, improving team throughput and documentation quality.

Technical Skills

RF & Microwave Design Tools:

- CST Microwave Studio (RF Simulation & Electromagnetic Analysis)
- Zuken CADSTAR (PCB Design & Layout)

Testing & Measurement:

- Spectrum Analyzer, Network Analyzer, Signal Generator, Power Meter, Oscilloscope, and Multimeter
- RF Test Automation and Performance Validation (5G, Wi-Fi 6E, Bluetooth)

Engineering & Analysis:

- Root Cause Analysis (4M, 5W1H, Fishbone Diagram)
- Process Improvement & Failure Analysis (FA)
- EMI/EMC Compliance & Troubleshooting
- Impedance Matching and RF Signal Integrity Analysis

Software & Programming:

- C Programming
- Python (Data Analysis, Automation, Report Generation)

Project Management & Documentation:

- Project Planning, Scheduling, and Execution
- Technical Documentation (ATP, QTP, QAP, ETP, TM, UM)
- Cross-Functional Team Collaboration & Technical Reporting

Core Competencies:

- RF/Microwave Circuit Design & Testing
- Power Amplifier, LNA, Mixer, and Filter Design
- System-Level Integration & Verification
- Process Capability Analysis (Cp, Cpk)
- Design of Experiments (DOE) and Data-Driven Decision Making

Education:

- **Bachelor of Technology (B.Tech) – Electronics and Communication Engineering**
Raghu Engineering College, Visakhapatnam, Andhra Pradesh
Graduated: January 2016 | GPA: 62.28%
- **Diploma – Electronics and Communication Engineering**
Bhaskar Polytechnic, Komati Palli, Bobbili, Andhra Pradesh
Completed: May 2013 | GPA: 75.85%

- **Secondary School Certificate (SSC)**
Sathya High School, Badangi, Andhra Pradesh
Completed: April 2008

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Declaration:

I hereby declare that the information provided in this resume is true and correct to the best of my knowledge and belief. I am fully aware that any false information or misrepresentation will lead to immediate termination of my employment.

Date:

Place: Bengaluru.

Signature

(P. Jagadeeswararao)
